Pragna Mamidipaka

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Hi! I am a PhD candidate specializing in Systems and Data Center Networking. In my current work, I focus on developing frameworks to make congestion control evaluations more representative, and hence better guide the design of high-performance network algorithms.

Education

Carnegie Mellon University — Pittsburgh, PA, USA
PhD in Electrical and Computer Engineering — *2024-2029*
Advisor : Prof. Theophilus Benson
CGPA : **4.00**/4.00

Indian Institute of Technology, Hyderabad — Hyderabad, India
Bachelor of Technology in Electrical Engineering, with minor in Artificial Intelligence — *2020 - 2024*
CGPA : **9.75**/10.00
President of India Gold Medallist

City International School, Pune — Pune, India
High School and Senior Secondary Education (Science Discipline) — *2007-2020*
Grade 12 : **95%** Grade 10 : **97%**

Research Areas and Publications

Data Center Congestion Control

Projects

Evaluating Congestion Control Under Incast Bursts — Oct 2025 – Present
with Prof. Theo Benson and Srikanth Sundaresan (Meta)

- A CCA-aware evaluation framework that maps incast burst parameters to key burst dimensions, enabling systematic coverage of congestion control behaviors under incast workloads.

Publications

- *Towards Truly Burst Aware Evaluation of Data Center Congestion Control* **(2026)**
  Asia-Pacific Workshop on Networking (APNet)

eBPF Management Abstractions

Projects

Prism: Towards Software Defined Observability — Feb 2025 – Oct 2025
with Prof. Theophilus Benson at CMU

- Introduces a reactive programming model for observability using eBPF probes by decoupling probe specification from execution.

Apiary: A Runtime Framework for Distributed eBPF Program Management — Aug 2024 – Oct 2025
with Prof. Theophilus Benson at CMU

- Proposes abstractions for distributed eBPF applications, addressing isolation and lifecycle management in large-scale deployments.

Leveraging Bytecode Understanding for Debugging eBPF NF Deployments — Sep 2023 – May 2024
with Dr. Praveen Tammana and Dr. Ramakrishna Upadrasta at IIT Hyderabad

- Built a system that analyzes packet behavior using only eBPF bytecode, enabling debugging without source code access.

eBPF Program Registry and Code Summarization — Apr 2023 – Dec 2023
with Dr. Palanivel Kodeswaran and Dr. Sayandeep Sen (IBM Research Lab)

- Developed an open-source registry with LLM-based summarization and ElasticSearch-powered semantic search for eBPF programs.

Publications

- *Yaksha-Prashna: Understanding eBPF Bytecode Network Function Behavior* **(2026)**
  arXiv Preprint
- *Apiary: Distributed Programming and Lifecycle Management for eBPF* **(2025)**
  N2Women Workshop @ SIGCOMM 2025

Pulsar Astronomy
Publications

- *The Indian Pulsar Timing Array Data Release 2: I. Dataset and Timing Analysis* **(2025)**
  Publications of the Astronomical Society of Australia
- *Low-frequency pulse-jitter measurement with the uGMRT I: PSR J0437–4715* **(2024)**
  Publications of the Astronomical Society of Australia
- *Application of Efron-Petrosian method to radio pulsar fluxes* **(2024)**
  Journal of Cosmology and Astroparticle Physics
- *Do Pulsar and Fast Radio Burst dispersion measures obey Benford's law?* **(2023)**
  Astroparticle Physics

Honors and Awards

Prabhu and Poonam Goel Graduate Fellowship — Mar 2026
Awarded by the CMU ECE Department.

Carnegie Institute of Technology Dean's Fellow — Aug 2024 - July 2025
Awarded full tuition support and stipend for one year of PhD study.

President of India Gold Medal — July 2024
For achieving the highest overall CGPA among all graduating B.Tech students at IIT Hyderabad.

Institute Silver Medal — July 2024
For achieving the highest CGPA in the Electrical Engineering Department at IIT Hyderabad.

Academic Excellence Award — Apr 2024, Apr 2022
For achieving the highest CGPA in Electrical Engineering Department at IIT Hyderabad, in the calendar years 2023 and 2021.

Excellence in Innovation Award — Jan 2022
Awarded for being one of top 5 among 1,300 participants in the Road2Shine program, a joint venture by the Indian and Japanese governments.

All India Rank of 3418 in JEE-Advanced — Nov 2020
among 1 Million Candidates taking the test.

Merit Certificate by CBSE — Apr 2018
For scoring in the top 0.1% among 1.7 Million candidates in Science in Grade 10.

Professional Service and Outreach Activities

Artifact Evaluation Committee Member, CAIS — 2026
Served as part of Artifact Evaluation Committee for the first ACM Conference on AI and Agentic Systems.

K-12 Community Outreach — 2026
Goals : Making engineering more accessible and equitable in K-12 spaces, re-envisioning engineering education pedagogy.

Shadow PC Member, IMC — 2026

Shadow PC Member, ACM CoNEXT — 2025

Evaluator, Meeting of the Minds – Carnegie Mellon University 2025
Reviewed senior undergraduate research projects as part of
CMU's annual research symposium.

Yoga Instructor at CMU 2024 - Present
Teach weekly group yoga sessions as part of CMU's GroupX fitness program.

Volunteer Teacher - National Service Scheme, India 2022-2024
Volunteered as a teacher and motivational lectures lead of Aksharamaala Program in India,
teaching Grade 10 Mathematics at rural schools.

Work Experience

Software Developer May 2023 – July 2023
Arcesium *Hyderabad, India*

Developed an application to analyze the CI/CD workflow in a Dev-Ops setting. This involved extensive work with APIs and dashboards like Apache Superset and Grafana.

Undergraduate Researcher Dec 2022
University of Tokyo *Tokyo, Japan*

Visited UTokyo for a research internship under the guidance of Prof. Kunihiko Sadakane. The project was based on succinctly encoding ordinal trees using tree covering.

Student Member Jun 2022 - Jun 2024
Indian Pulsar Timing Array *Hyderabad, India*

I was a student member of the InPTA Consortium, as part of the Data Reduction team. I also worked with National Centre for Radio Astrophysics for collating legacy data of the uGMRT (Giant Metrewave Radio Telescope).

5G Testbed Intern Jan 2022 – Jun 2022
Wisig Networks *Hyderabad, India*

Developed a module to transmit DCI (Downlink Control Information) from gNB Base station to User equipment. Gained knowledge about L2/L3 protocol stack development.